



SIMATS Online Programming Lab

JAVA PROGRAMMING



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CO3-AT3-Assignment

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QUESTIONS

Q1

Bank Account ♦ Concurrent
Deposit and Withdrawal
25 marks

**Q2**

Traffic Signal Simulation ♦
Thread Control
25 marks

**Q3**

Thread Priority ♦ Hospital
Emergency Tasks
25 marks

**Q4**

Online Login System ♦
Multiple Exceptions
25 marks



4/4 answered

Q2

Traffic Signal Simulation ♦ Thread Control

25
marks

Problem Statement:

Develop a Java program to simulate traffic signals using separate threads for Red Yellow and Green signals.

Requirements:

Create separate threads for each signal.

Use sleep() to control the duration of each signal.

Display the signal status sequentially.

Demonstrate thread execution and termination.

Handle any relevant exceptions using try-catch.

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.println("Enter signal: 1-
9         int choice = sc.nextInt();
10
11         Runnable signal = new Runnable() {
```

```
12         public void run() {
13             try {
14                 if (choice == 1) {
15                     System.out.println("
16                     Thread.sleep(3000);
17                     System.out.println("
18                 }
19                 else if (choice == 2) {
20                     System.out.println("
21                     Thread.sleep(2000);
22                     System.out.println("
23                 }
24                 else if (choice == 3) {
25                     System.out.println("
26                     Thread.sleep(3000);
27                     System.out.println("
28                 }
29                 else {
30                     System.out.println("
31                 }
32             } catch (InterruptedException e) {
33                 System.out.println("Thre
34             }
35         }
36     };
37
38     Thread t = new Thread(signal);
39     t.start();
40
41     try {
42         t.join();
43     } catch (InterruptedException e) {
44         System.out.println("Main thread
45     }
46
47     System.out.println("Traffic signal c
48 }
49 }
```

Input

1

Output

 Run

 Save

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